

Car Colour Detection and Traffic Analysis

1) Project Overview

This project extends the Age-Gender Detection training project by implementing a Vehicle Colour Detection and Traffic Analysis System.

The system combines Deep Learning and Computer Vision techniques to:

1. Detect vehicles in traffic scenes.
2. Predict vehicle colours.
3. Count the number of vehicles present.
4. Count the number of pedestrians present.
5. Highlight blue vehicles with red bounding boxes.
6. Highlight vehicles of other colours with blue bounding boxes.
7. Provide a graphical user interface (GUI) for image upload and result visualization.

2) Problem Statement

Traffic monitoring systems require automated vehicle analysis for smart city and intelligent transportation applications.

The objective of this project is to develop a machine learning system capable of:

1. Detecting vehicles and pedestrians in traffic images.
2. Predicting the colour of detected vehicles.
3. Counting vehicles and pedestrians.
4. Displaying results through a user-friendly GUI.

3) Dataset

Vehicle Colour Recognition Dataset (VCoR)

Dataset Source:

Kaggle – Vehicle Colour Recognition Dataset

Vehicle colour classes include:

- Black
- Blue
- Brown
- Green
- Grey
- Orange
- Red

- Silver
- White
- Yellow

Dataset Structure:

train/
validation/
test/

Images were organized into class-specific folders for supervised learning.

4) Methodology

1. Data Preprocessing

- Image resizing to 224×224 pixels
- Image normalization
- Batch loading using TensorFlow Dataset API

2. Vehicle Colour Classification Model

A Convolutional Neural Network (CNN) was trained to classify vehicle colours.

Model Architecture:

- Convolution Layer
- Max Pooling Layer
- Convolution Layer
- Max Pooling Layer
- Dense Layer
- Softmax Output Layer

3. Vehicle and Pedestrian Detection

YOLOv8 was used for object detection.

Detected Classes:

- Car
- Bus
- Truck
- Motorcycle
- Person

4. Traffic Analysis Logic

For each detected vehicle:

1. Crop vehicle region.
2. Predict vehicle colour using CNN.
3. If colour is Blue:
 - Draw Red Rectangle.
4. Else:
 - Draw Blue Rectangle.

Pedestrians are counted separately.

5) GUI Development

Gradio was used to develop a graphical user interface.

Features:

- Upload traffic image
- Preview image
- Detect vehicles
- Predict vehicle colours
- Count pedestrians
- Display annotated output image
- Display summary statistics

6) Results

Vehicle Colour Classification

Test Accuracy: 78.86 %

Traffic Detection

Successfully detected:

- Vehicles
- Pedestrians
- Vehicle colours

Colour-Based Highlighting

- Blue Vehicles → Red Bounding Box
- Other Vehicles → Blue Bounding Box

7) Visualizations

Included in notebook/repository:

- Training Accuracy Graph
- Validation Accuracy Graph
- Training Loss Graph
- Validation Loss Graph
- Confusion Matrix
- Sample Predictions
- YOLO Detection Results

8) Technologies Used

- Python
- TensorFlow / Keras
- OpenCV
- NumPy
- Matplotlib
- Scikit-Learn
- YOLOv8
- Gradio

9) Future Improvements

- Real-time video processing
- Traffic density estimation
- Multi-camera integration
- Vehicle tracking
- Traffic signal optimization
- Deployment using Streamlit or Flask

10) Conclusion

This project successfully combines Deep Learning and Computer Vision techniques to perform vehicle colour recognition and traffic analysis. The developed system detects vehicles and pedestrians, predicts vehicle colours, counts traffic participants, and visualizes results through an interactive GUI.